

SIMATS ENGINEERING

ASSIGNMENT 1

COURSE: MACHINE LEARNING (ITA 0603)

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Unit I – Learning Problems – Perspectives and Issues - Concept Learning - Version Spaces& Candidate Elimination - Inductive Bias - Decision Tree Learning - Representation Algorithm- Heuristic Space Search

Design of a Machine Learning Model to Predict Student Academic Performance

1.Problem Understanding

Objective: To design and build a Machine Learning model that predicts a student's academic performance (final grade / marks, or a Pass–Fail outcome) using measurable academic and behavioural indicators, so that educators can identify at-risk students early and provide timely support.

Problem Statement:

Given historical student data — such as attendance percentage, study hours, previous internal marks, participation in extracurricular activities, and socio-economic indicators — predict the student's final academic score (regression) or whether the student will Pass/Fail or fall into a performance category such as High/Medium/Low (classification).

Key Objectives:

- Identify the most influential factors affecting student performance (e.g., attendance, study hours, prior grades).
- Build a predictive model that estimates a student's expected academic outcome with acceptable accuracy.
- Enable early identification of at-risk students so that timely academic intervention can be provided.
- Present findings through clear visualizations and a well-documented, reproducible workflow.

2.Data Collection & Preparation

Dataset:

A structured dataset such as the publicly available 'Student Performance Data Set' (UCI Machine Learning Repository) or an institution's own academic records is used. Each row represents one student, and columns represent the features listed below.

Sample Features:

Feature	Description	Type
Attendance (%)	Percentage of classes attended	Numeric
Study Hours/Week	Average hours spent studying per week	Numeric
Previous Grade	Marks obtained in the previous semester/exam	Numeric
Extracurricular	Participation in extracurricular activities	Categorical (Yes/No)
Parental Education	Highest education level of parents	Categorical
Internet Access	Availability of internet at home	Categorical (Yes/No)
Final Grade	Target variable to be predicted	Numeric Categorical

Preprocessing Steps:

- Handling missing values — numeric columns imputed with mean/median; categorical columns imputed with mode.
- Encoding categorical variables (e.g., Yes/No, gender, parental education) using Label Encoding or One-Hot Encoding.
- Feature scaling — standardizing numeric features (e.g., study hours, attendance) using StandardScaler so that no single feature dominates due to its scale.
- Removing duplicate records and detecting/handling outliers (e.g., attendance values outside 0–100%).
- Splitting the dataset into training (80%) and testing (20%) sets using `train_test_split` for unbiased evaluation.

Sample Preprocessing Code:

```
import pandas as pd
import numpy as np

df = pd.read_csv("student_performance.csv")
print(df.shape)
print(df.isnull().sum())

# Handle missing values
df.fillna(df.mean(numeric_only=True), inplace=True)

# Encode categorical columns
df['extracurricular'] = df['extracurricular'].map({'Yes': 1, 'No': 0})
df['internet_access'] = df['internet_access'].map({'Yes': 1, 'No': 0})

# Train-test split
from sklearn.model_selection import train_test_split
X = df.drop('final_grade', axis=1)
y = df['final_grade']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

3. Analysis Using Numpy/Pandas

Pandas is used to explore the dataset's structure, summary statistics, and relationships between features, while NumPy supports the underlying numerical computations (correlation, aggregation, array operations).

```
# Descriptive statistics
print(df.describe())
```

```
# Correlation of each feature with the target
correlation = df.corr(numeric_only=True)['final_grade'].sort_values(ascending=False)
print(correlation)
```

```
# Group-wise average performance
print(df.groupby('extracurricular')['final_grade'].mean())
```

```
# NumPy-based summary
grades = df['final_grade'].to_numpy()
print("Mean:", np.mean(grades), "Std Dev:", np.std(grades))
```

Expected Insight:

This analysis typically reveals that 'Previous Grade' and 'Attendance (%)' have the strongest positive correlation with 'Final Grade', followed by 'Study Hours/Week', while factors like extracurricular participation show a smaller but positive association — guiding which features are most important to retain for modeling.

4. Model Selection & Training

Since the target (final grade) is continuous, a regression approach is used. A Random Forest Regressor is selected for its ability to model non-linear relationships and rank feature importance, alongside a simpler Linear Regression model used as a baseline for comparison. (If the target is instead framed as Pass/Fail, the same pipeline can use Logistic Regression or a Random Forest Classifier.)

```
from sklearn.linear_model import LinearRegression
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
```

```
# Baseline model
lr = LinearRegression()
lr.fit(X_train, y_train)
lr_pred = lr.predict(X_test)
```

```
# Ensemble model
rf = RandomForestRegressor(n_estimators=200, random_state=42)
rf.fit(X_train, y_train)
rf_pred = rf.predict(X_test)
```

```
print("Linear Regression R2:", r2_score(y_test, lr_pred))
print("Random Forest R2:", r2_score(y_test, rf_pred))
```

```
print("Random Forest MAE:", mean_absolute_error(y_test, rf_pred))
```

Model Evaluation:

Metric	Purpose
R ² Score	Proportion of variance in final grade explained by the model (closer to 1 is better).
MAE (Mean Absolute Error)	Average absolute difference between predicted and actual grades.
RMSE (Root Mean Squared Error)	Penalizes larger prediction errors more heavily than MAE.

The Random Forest model is expected to outperform plain Linear Regression by capturing non-linear interactions (e.g., the combined effect of low attendance with low study hours), while remaining interpretable through its feature-importance scores.

5. Visualization & Interpretation

```
import matplotlib.pyplot as plt
import seaborn as sns
```

```
# Correlation heatmap
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap='coolwarm')
plt.title("Correlation Heatmap of Student Features")
plt.show()
```

```
# Study hours vs final grade
sns.scatterplot(x='study_hours', y='final_grade', data=df)
plt.title("Study Hours vs Final Grade")
plt.show()
```

```
# Feature importance from Random Forest
importances = pd.Series(rf.feature_importances_, index=X.columns).sort_values(ascending=False)
importances.plot(kind='bar', color='steelblue')
plt.title("Feature Importance for Predicting Final Grade")
plt.ylabel("Importance Score")

plt.show()
```

Interpretation of Visuals:

- Correlation Heatmap – visually confirms that Previous Grade and Attendance have the darkest (strongest) correlation with Final Grade, guiding feature prioritization.
 - Scatter Plot (Study Hours vs Final Grade) – shows a clear upward trend, confirming that increased study time is generally associated with better academic outcomes, with diminishing returns beyond a certain point.
 - Feature Importance Bar Chart – ranks which inputs the Random Forest model relies on most, offering an interpretable summary that teachers/administrators can act on (e.g., prioritizing attendance interventions).
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6. Innovation & SDG Mapping

This project directly aligns with United Nations Sustainable Development Goal 4 (SDG 4: Quality Education), which aims to 'ensure inclusive and equitable quality education and promote lifelong learning opportunities for all'.

Real-World Relevance:

- Early Warning System – the model can flag students at risk of underperforming well before final exams, enabling teachers to intervene with remedial classes or mentoring.
- Personalized Learning – predictions can be used to recommend individualized study plans based on each student's influential factors (e.g., more structured study time for students with low prior scores).
- Reducing Dropout Rates – by identifying at-risk students early, institutions can take proactive steps that support SDG 4's target of reducing school dropout and improving completion rates.
- Equitable Support – highlighting socio-economic or access-related factors (e.g., internet access) helps institutions direct resources to underserved students, supporting inclusive and equitable education.

Innovation Aspect: Combining predictive analytics with an interpretable feature-importance report turns a black-box prediction into an actionable diagnostic tool for educators — moving beyond simple grade prediction toward genuine, data-driven academic support.

7. Documentation & Conclusion

Summary of Workflow:

Data Collection → Preprocessing (cleaning, encoding, scaling) → Exploratory Analysis (Pandas/NumPy) → Visualization (Matplotlib/Seaborn) → Model Training (Linear Regression & Random Forest) → Evaluation (R^2 , MAE, RMSE) → Insight Generation & SDG Mapping.

Conclusion:

The designed model demonstrates that a student's academic performance can be reasonably predicted from measurable indicators such as attendance, prior grades, and study hours, with the Random Forest model expected to provide the strongest predictive accuracy due to its ability to capture non-linear feature interactions. Beyond prediction accuracy, the project's real value lies in its ability to surface actionable insights — identifying which factors matter most — so that educators can intervene early, personalize support, and work toward more equitable and effective learning outcomes, directly supporting SDG 4: Quality Education.

Future Scope:

- Incorporate additional behavioural data (e.g., LMS engagement logs) to further improve prediction accuracy.
- Deploy the model as a dashboard for faculty to monitor at-risk students in real time each semester.
- Experiment with advanced models such as Gradient Boosting (XGBoost) or a small neural network for potentially higher accuracy.